

# Phase 0: Vector Math Foundation

## *The Language of Electromagnetic Fields*

Master Study Guide & Step-by-Step Derivations

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# 1 Introduction: Why Vector Calculus?

In standard circuit theory, you deal with *lumped* parameters like Voltage ( $V$ ) and Current ( $I$ ). These are scalars. You can measure them with a multimeter at a single point on a wire.

However, in Electromagnetic Field Theory, energy exists in the empty space *between* the wires. This energy is distributed continuously. To describe how a magnetic field bends around a wire, or how an electric field radiates outward from an antenna, we cannot use simple numbers. We must use **Vector Fields**.

A vector field assigns a magnitude and a direction to every single point in 3D space. To analyze how these fields change, stretch, or rotate as we move through space, we need a specialized mathematical language: **Vector Calculus**.

## 2 The Stage: 3D Coordinate Systems

Before we can do math, we must define the "stage" upon which our fields exist. Depending on the physical geometry of the problem (e.g., a flat plate vs. a round wire vs. a spherical charge), choosing the right coordinate system turns an impossible integral into a simple one.

### 2.1 1. Cartesian Coordinates ( $x, y, z$ )

**What it is:** The standard grid system. You define a point by walking along the x-axis, then the y-axis, then the z-axis.

**When to use it:** When dealing with flat planes, rectangular waveguides, or infinite straight sheets of charge.

**Differential Elements:** To do calculus, we must know the size of an infinitesimally small "chunk" of space.

- **Differential Length:**  $d\mathbf{l} = dx\hat{\mathbf{i}} + dy\hat{\mathbf{j}} + dz\hat{\mathbf{k}}$
- **Differential Volume:**  $dv = dx\,dy\,dz$

### 2.2 2. Cylindrical Coordinates ( $r, \phi, z$ )

**What it is:** Imagine a tin can. To find a point, you walk out from the center a radial distance  $r$ , rotate around the center by an angle  $\phi$ , and then move straight up to a height  $z$ .

**When to use it:** When dealing with coaxial cables, infinite straight wires, or circular loops.

**Scaffolded Logic for Differential Length:** If you increase  $r$  by a tiny amount  $dr$ , you move a distance  $dr$ . If you increase  $z$  by a tiny amount  $dz$ , you move a distance  $dz$ . But if you increase the angle  $\phi$  by a tiny amount  $d\phi$ , the physical distance you travel depends on how far out you are from the center. From basic geometry, arc length equals radius times angle. So, the distance is  $rd\phi$ .

**Differential Elements:**

- **Differential Length:**  $d\mathbf{l} = dr\hat{\mathbf{r}} + rd\phi\hat{\phi} + dz\hat{\mathbf{z}}$
- **Differential Volume:**  $dv = (dr) \cdot (rd\phi) \cdot (dz) = r dr d\phi dz$

## 2.3 3. Spherical Coordinates ( $r, \theta, \phi$ )

**What it is:** Imagine the Earth. To find a point, you specify the distance from the core ( $r$ ), the latitude angle dropping down from the North Pole ( $\theta$ ), and the longitude angle spinning around the equator ( $\phi$ ).

**When to use it:** When dealing with point charges, spherical capacitors, or antenna radiation patterns.

**Differential Elements:** Using the same arc-length logic as above:

- **Differential Length:**  $d\mathbf{l} = dr\hat{\mathbf{r}} + rd\theta\hat{\theta} + r\sin\theta d\phi\hat{\phi}$
- **Differential Volume:**  $dv = (dr) \cdot (rd\theta) \cdot (r\sin\theta d\phi) = r^2 \sin\theta dr d\theta d\phi$

## 3 The Slope: The Del Operator and Gradient

### 3.1 The Del Operator ( $\nabla$ )

In 1D calculus, the derivative  $\frac{d}{dx}$  tells you the slope of a line. In 3D space, things can change in the x, y, and z directions simultaneously.

To handle this, we define the **Del Operator** ( $\nabla$ ). It is a "vector derivative". It doesn't mean anything on its own until you apply it to a field.

In Cartesian coordinates, it is defined as:

$$\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \quad (1)$$

### 3.2 The Gradient ( $\nabla V$ )

**The Concept (Building from knowns):** Imagine you are standing on the side of a mountain. The altitude of the mountain is a scalar field  $V(x, y)$ . At any point, there is a specific direction you can walk to go uphill the fastest.

If you apply the Del operator directly to a scalar field, you get the **Gradient**. The gradient produces a vector that points in the direction of the *maximum rate of increase* of that scalar field.

$$\nabla V = \frac{\partial V}{\partial x} \hat{\mathbf{i}} + \frac{\partial V}{\partial y} \hat{\mathbf{j}} + \frac{\partial V}{\partial z} \hat{\mathbf{k}} \quad (2)$$

**Practical Justification (Why do we care?):** In Phase 1, you will learn that charges create a scalar "Electric Potential" field ( $V$ ). A positive charge naturally wants to roll "downhill" from high potential to low potential. Therefore, the Electric Field ( $\mathbf{E}$ ) is simply the negative gradient of the potential field!

$$\mathbf{E} = -\nabla V \quad (3)$$

## 4 The Source: Divergence and the Divergence Theorem

### 4.1 The Concept of Divergence ( $\nabla \cdot \mathbf{A}$ )

**The Concept (Building from knowns):** Imagine a vector field representing the flow of water in a river. If you place a tiny, invisible cube in the water, water flows in one side and out the other.

If more water flows OUT of the cube than flows IN, that specific point in space must contain a hidden faucet (a source). If more water flows IN than OUT, it must contain a hidden drain (a sink).

The **Divergence** of a vector field mathematically measures exactly this: the net outward flow (flux) per unit volume at a specific point. Because we are taking the dot product of the vector operator with a vector field, the result is a **scalar**.

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (4)$$

**Practical Justification (Why do we care?):** In Phase 1, you will learn Gauss's Law. Electric flux lines "diverge" outward from positive charges. Therefore, the divergence of the electric flux density field ( $\mathbf{D}$ ) exactly equals the volume charge density ( $\rho_v$ ) hiding at that point!

$$\nabla \cdot \mathbf{D} = \rho_v \quad (5)$$

### 4.2 PYQ Derivation: Divergence in Cylindrical Coordinates

*Note: This is a highly tested exam question. You must know how to derive the divergence formula in cylindrical coordinates from first principles, without just memorizing the final formula.*

#### Step 1: Define the Vector Field and the Volume Element

Let an arbitrary vector field be expressed in cylindrical coordinates  $(r, \phi, z)$  as:

$$\mathbf{A} = A_r \hat{\mathbf{r}} + A_\phi \hat{\boldsymbol{\phi}} + A_z \hat{\mathbf{z}} \quad (6)$$

We consider an infinitesimally small "wedge" of volume  $\Delta v$  located at point  $P(r, \phi, z)$ . The dimensions of this differential wedge are:

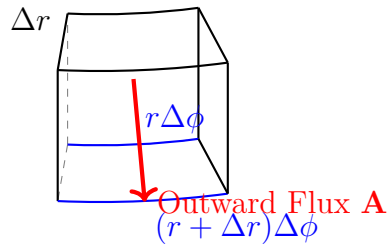
- Radial length:  $\Delta r$

- Arc length (inner):  $r\Delta\phi$
- Arc length (outer):  $(r + \Delta r)\Delta\phi$
- Vertical height:  $\Delta z$

The total volume of this tiny wedge is simply length  $\times$  width  $\times$  height:

$$\Delta v = r \Delta r \Delta\phi \Delta z \quad (7)$$

### TikZ Diagram: Cylindrical Differential Element



### Step 2: Calculate Flux through the Radial ( $r$ ) Faces

The net outward flux in the radial direction ( $\Psi_r$ ) is the flux leaving the outer front face MINUS the flux entering the inner back face.

Flux = (Vector Field Component)  $\times$  (Area of the Face)

*Crucial Deduction:* In Cartesian coordinates, the front and back faces of a cube are exactly the same size. However, look at the diagram above. In cylindrical coordinates, as you move further from the center, the arc length grows. The front face is physically larger than the back face!

Area of inner back face (at radius  $r$ ):

$$S_{\text{inner}} = (r\Delta\phi)\Delta z \quad (8)$$

Area of outer front face (at radius  $r + \Delta r$ ):

$$S_{\text{outer}} = ((r + \Delta r)\Delta\phi)\Delta z \quad (9)$$

The flux entering the back face is exactly evaluated at  $r$ :

$$\Psi_{\text{in}} = A_r(r) \cdot (r\Delta\phi\Delta z) \quad (10)$$

The flux leaving the front face is evaluated at  $r + \Delta r$ . We use a first-order Taylor series to approximate the value of the function  $(rA_r)$  as we move a small distance  $\Delta r$ :

$$f(r + \Delta r) \approx f(r) + \frac{\partial f}{\partial r} \Delta r \quad (11)$$

Applying this to our flux term:

$$\Psi_{\text{out}} = A_r(r + \Delta r) \cdot ((r + \Delta r)\Delta\phi\Delta z) \approx \left[ rA_r + \frac{\partial}{\partial r}(rA_r)\Delta r \right] \Delta\phi\Delta z \quad (12)$$

Subtracting the two to find the net outward radial flux:

$$\Psi_r = \Psi_{\text{out}} - \Psi_{\text{in}} \quad (13)$$

$$\Psi_r = \left[ rA_r + \frac{\partial}{\partial r}(rA_r)\Delta r \right] \Delta\phi\Delta z - [rA_r] \Delta\phi\Delta z \quad (14)$$

$$\Psi_r = \frac{\partial}{\partial r}(rA_r)\Delta r\Delta\phi\Delta z \quad (15)$$

### Step 3: Calculate Flux through the Azimuthal ( $\phi$ ) Faces

The left and right faces are located at angle  $\phi$  and  $\phi + \Delta\phi$ . Unlike the radial direction, the area of these two faces is identical:  $\Delta r\Delta z$ .

Using the same Taylor series logic for a small step  $\Delta\phi$ :

$$\Psi_\phi = \Psi_{\text{right}} - \Psi_{\text{left}} \quad (16)$$

$$\Psi_\phi = \left[ A_\phi + \frac{\partial A_\phi}{\partial \phi}\Delta\phi \right] \Delta r\Delta z - [A_\phi]\Delta r\Delta z \quad (17)$$

$$\Psi_\phi = \frac{\partial A_\phi}{\partial \phi}\Delta r\Delta\phi\Delta z \quad (18)$$

### Step 4: Calculate Flux through the Axial ( $z$ ) Faces

The top and bottom faces are located at height  $z$  and  $z + \Delta z$ . The area of these faces is also identical:  $r\Delta r\Delta\phi$ .

Again, using the Taylor series logic for a small step  $\Delta z$ :

$$\Psi_z = \Psi_{\text{top}} - \Psi_{\text{bottom}} \quad (19)$$

$$\Psi_z = \left[ A_z + \frac{\partial A_z}{\partial z}\Delta z \right] r\Delta r\Delta\phi - [A_z]r\Delta r\Delta\phi \quad (20)$$

$$\Psi_z = \frac{\partial A_z}{\partial z}r\Delta r\Delta\phi\Delta z \quad (21)$$

### Step 5: Assemble the Final Divergence Formula

The fundamental mathematical definition of divergence is the total net outward flux divided by the differential volume, as the volume shrinks to zero:

$$\nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\Psi_r + \Psi_\phi + \Psi_z}{\Delta v} \quad (22)$$

Recall our total volume was  $\Delta v = r\Delta r\Delta\phi\Delta z$ . We substitute all our derived terms into the numerator:

$$\nabla \cdot \mathbf{A} = \frac{\left(\frac{\partial}{\partial r}(rA_r)\Delta r\Delta\phi\Delta z\right) + \left(\frac{\partial A_\phi}{\partial\phi}\Delta r\Delta\phi\Delta z\right) + \left(\frac{\partial A_z}{\partial z}r\Delta r\Delta\phi\Delta z\right)}{r\Delta r\Delta\phi\Delta z} \quad (23)$$

Notice the beautiful cancellation. We can divide each of the three terms in the numerator individually by the denominator. All the differential lengths ( $\Delta r, \Delta\phi, \Delta z$ ) cancel out perfectly!

$$\text{Term 1: } \frac{\frac{\partial}{\partial r}(rA_r)\Delta r\Delta\phi\Delta z}{r\Delta r\Delta\phi\Delta z} = \frac{1}{r} \frac{\partial}{\partial r}(rA_r) \quad (24)$$

$$\text{Term 2: } \frac{\frac{\partial A_\phi}{\partial\phi}\Delta r\Delta\phi\Delta z}{r\Delta r\Delta\phi\Delta z} = \frac{1}{r} \frac{\partial A_\phi}{\partial\phi} \quad (25)$$

$$\text{Term 3: } \frac{\frac{\partial A_z}{\partial z}r\Delta r\Delta\phi\Delta z}{r\Delta r\Delta\phi\Delta z} = \frac{\partial A_z}{\partial z} \quad (26)$$

Adding these simplified terms together yields the final, rigorously proven formula for divergence in cylindrical coordinates.

**Final Derived Formula:**

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r}(rA_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial\phi} + \frac{\partial A_z}{\partial z} \quad (27)$$

### 4.3 The Divergence Theorem (Gauss's Theorem)

**The Concept (Bridging the Local to the Global):** We just spent a lot of time defining divergence as the flux leaving a microscopic, invisibly small cube. But engineers build real, macro-sized objects like antennas and capacitors. How do we scale this up?

Imagine gluing billions of these tiny microscopic cubes together to form a large balloon (a macroscopic volume  $V$ ). Every time water (flux) flows OUT of the right face of one tiny cube, it immediately flows IN to the left face of the neighboring cube. The internal fluxes perfectly cancel each other out everywhere!

The only flux that does *not* cancel out is the flux located on the very outer skin of the balloon, because there is no neighboring cube there to absorb it.



Therefore, if you mathematically sum up (integrate) all the tiny microscopic sources (divergences) inside the volume, it must exactly equal the total macroscopic flux escaping through the outer boundary surface  $S$ .

**The Theorem Statement:** The volume integral of the divergence of a vector field over any volume  $V$  is identically equal to the closed surface integral of that vector field over the surface  $S$  bounding that volume.

$$\iiint_V (\nabla \cdot \mathbf{A}) dv = \oint_S \mathbf{A} \cdot d\mathbf{S} \quad (28)$$

**Why it works for exams:** This theorem is a magic trick. If an exam asks you to calculate a brutal surface integral over six faces of a cube, do not do it! Use the Divergence Theorem to convert it into a simple volume integral of a scalar divergence instead. It takes 3 lines of math instead of 3 pages.

## 5 The Rotation: Curl and Stokes' Theorem

### 5.1 The Concept of Curl ( $\nabla \times \mathbf{A}$ )

**The Concept (Building from knowns):** Imagine our river again. Last time we looked for hidden faucets. Now, imagine you drop a tiny, frictionless paddle-wheel into the water.

If the water on the right side of the paddle is flowing faster than the water on the left side, the paddle-wheel will begin to spin.

The **Curl** of a vector field mathematically measures this exact phenomenon: the localized "vorticity" or macroscopic circulation per unit area at a specific point. Because rotation inherently has an axis (the stick holding the paddle-wheel), the Curl produces a **Vector** quantity. The vector points along the axis of rotation, defined by the right-hand rule.

$$\nabla \times \mathbf{A} = \left( \lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \hat{\mathbf{n}} \quad (29)$$

**Practical Justification (Why do we care?):** In Phase 4, you will learn Ampere's Law. Physical electrical current ( $\mathbf{J}$ ) creates magnetic fields ( $\mathbf{H}$ ) that wrap around the wire in continuous circles. Therefore, current acts as the source of magnetic "curl".

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (30)$$

Furthermore, in Phase 5, Faraday's Law reveals that a changing magnetic field literally "curls" an electric field into existence!

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (31)$$

### 5.2 PYQ Derivation: Curl in Cartesian Coordinates

*Note: This is another highly tested, mandatory derivation for university exams. It requires calculating line integrals around a differential square.*

**Step 1: Define the Vector Field and the Area Element**

Let an arbitrary vector field be expressed in Cartesian coordinates  $(x, y, z)$  as:

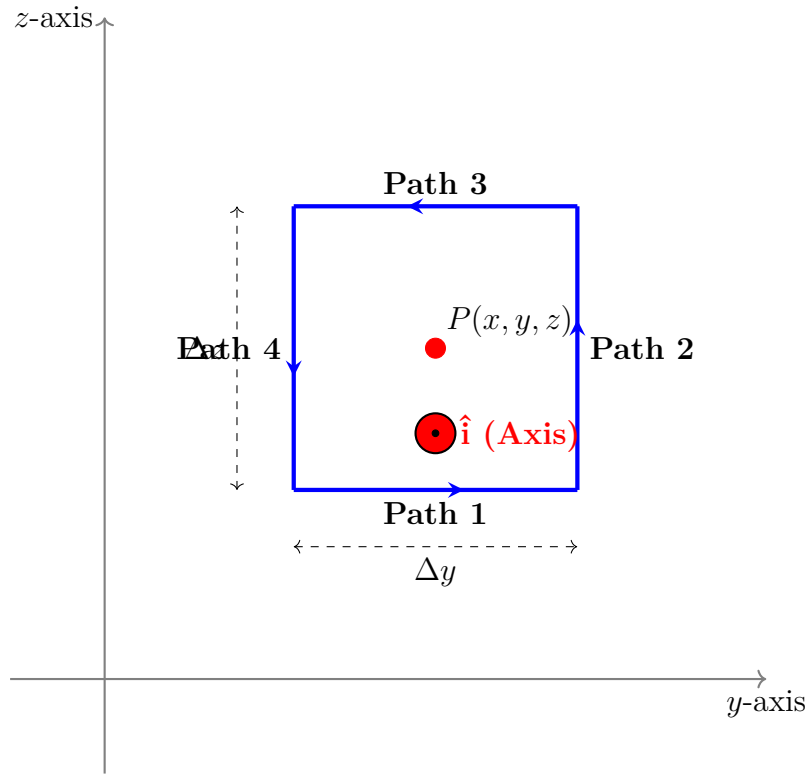
$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}} \quad (32)$$

Because Curl is a 3D vector, we must calculate its x, y, and z components separately. Let us focus exclusively on finding the **x-component** of the curl.

By the right-hand rule, rotation around the x-axis occurs purely in the  $y - z$  plane. Therefore, we define a microscopic, rectangular loop situated perfectly in the  $y - z$  plane, centered at point  $P(x, y, z)$ . The dimensions of this loop are:

- Width:  $\Delta y$
- Height:  $\Delta z$
- Area:  $\Delta S_x = \Delta y \Delta z$

### TikZ Diagram: Cartesian Differential Circulation Loop



### Step 2: Evaluate the Four Line Integrals

We must compute the line integral  $\oint \mathbf{A} \cdot d\mathbf{l}$  sequentially around the four defined paths in a strict counter-clockwise direction (to adhere to the right-hand rule).

**Path 1 (Bottom Edge):** We travel from left to right. The displacement is purely in the positive y-direction, so  $d\mathbf{l} = dy\hat{\mathbf{j}}$ . Since the path is located vertically below the center by a distance  $\Delta z/2$ , we approximate the vector field along this line using a first-order Taylor series:

$$A_{y,\text{bottom}} \approx A_y(x, y, z) - \frac{\partial A_y}{\partial z} \left( \frac{\Delta z}{2} \right) \quad (33)$$

The line integral contribution from Path 1 is:

$$I_1 = \int_{\text{Path 1}} \mathbf{A} \cdot d\mathbf{l} = \left[ A_y - \frac{\partial A_y}{\partial z} \frac{\Delta z}{2} \right] \Delta y \quad (34)$$

**Path 2 (Right Edge):** We travel upward. The displacement is purely in the positive z-direction, so  $d\mathbf{l} = dz\hat{\mathbf{k}}$ . This path is located to the right of the center by a distance  $\Delta y/2$ . The field approximation is:

$$A_{z,\text{right}} \approx A_z(x, y, z) + \frac{\partial A_z}{\partial y} \left( \frac{\Delta y}{2} \right) \quad (35)$$

The line integral contribution from Path 2 is:

$$I_2 = \int_{\text{Path 2}} \mathbf{A} \cdot d\mathbf{l} = \left[ A_z + \frac{\partial A_z}{\partial y} \frac{\Delta y}{2} \right] \Delta z \quad (36)$$

**Path 3 (Top Edge):** We travel backward from right to left. The displacement is in the *negative* y-direction, so  $d\mathbf{l} = -dy\hat{\mathbf{j}}$ . Note the crucial negative sign! This path is located vertically above the center by  $\Delta z/2$ .

$$A_{y,\text{top}} \approx A_y(x, y, z) + \frac{\partial A_y}{\partial z} \left( \frac{\Delta z}{2} \right) \quad (37)$$

The line integral contribution from Path 3 is:

$$I_3 = \int_{\text{Path 3}} \mathbf{A} \cdot d\mathbf{l} = - \left[ A_y + \frac{\partial A_y}{\partial z} \frac{\Delta z}{2} \right] \Delta y \quad (38)$$

**Path 4 (Left Edge):** We travel downward. The displacement is in the *negative* z-direction, so  $d\mathbf{l} = -dz\hat{\mathbf{k}}$ . This path is located to the left of the center by  $\Delta y/2$ .

$$A_{z,\text{left}} \approx A_z(x, y, z) - \frac{\partial A_z}{\partial y} \left( \frac{\Delta y}{2} \right) \quad (39)$$

The line integral contribution from Path 4 is:

$$I_4 = \int_{\text{Path 4}} \mathbf{A} \cdot d\mathbf{l} = - \left[ A_z - \frac{\partial A_z}{\partial y} \frac{\Delta y}{2} \right] \Delta z \quad (40)$$

### Step 3: Sum the Integrals and Divide by Area

The total macroscopic circulation around the entire loop is the direct sum of the four

segments:

$$\oint_{\text{Loop}} \mathbf{A} \cdot d\mathbf{l} = I_1 + I_2 + I_3 + I_4 \quad (41)$$

Let us carefully expand and sum these terms:

$$\begin{aligned} \oint \mathbf{A} \cdot d\mathbf{l} &= \left( A_y \Delta y - \frac{\Delta y \Delta z}{2} \frac{\partial A_y}{\partial z} \right) \\ &+ \left( A_z \Delta z + \frac{\Delta y \Delta z}{2} \frac{\partial A_z}{\partial y} \right) \\ &- \left( A_y \Delta y + \frac{\Delta y \Delta z}{2} \frac{\partial A_y}{\partial z} \right) \\ &- \left( A_z \Delta z - \frac{\Delta y \Delta z}{2} \frac{\partial A_z}{\partial y} \right) \end{aligned} \quad (42)$$

*Logical Deduction:* Notice that the base field values ( $A_y \Delta y$  and  $A_z \Delta z$ ) completely and perfectly cancel out! The uniform, straight-flowing parts of the field contribute zero to the total rotation. Only the spatial derivative terms (the *changes* in the field) survive.

Grouping the surviving derivative terms together:

$$\oint \mathbf{A} \cdot d\mathbf{l} = \left( \frac{1}{2} \frac{\partial A_z}{\partial y} + \frac{1}{2} \frac{\partial A_z}{\partial y} \right) \Delta y \Delta z - \left( \frac{1}{2} \frac{\partial A_y}{\partial z} + \frac{1}{2} \frac{\partial A_y}{\partial z} \right) \Delta y \Delta z \quad (43)$$

$$\oint \mathbf{A} \cdot d\mathbf{l} = \left( \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \Delta y \Delta z \quad (44)$$

By the fundamental definition of the curl, we divide this total circulation by the area of our loop ( $\Delta S_x = \Delta y \Delta z$ ):

$$(\nabla \times \mathbf{A})_x = \frac{\oint \mathbf{A} \cdot d\mathbf{l}}{\Delta y \Delta z} = \frac{\left( \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \Delta y \Delta z}{\Delta y \Delta z} \quad (45)$$

The area terms flawlessly cancel, leaving us with the exact, rigorous expression for the x-component of the Cartesian curl:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \quad (46)$$

By repeating this exact cyclical geometric derivation for a loop placed in the  $x - z$  plane, and another loop placed in the  $x - y$  plane, we derive the remaining  $y$  and  $z$  components.

**Final Derived Formula:** The full 3D vector expression is beautifully and conve-

niently summarized as a matrix determinant cross product:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (47)$$

### 5.3 Stokes' Theorem

**The Concept (Bridging the Local to the Global):** Much like the Divergence Theorem, Stokes' Theorem solves a major scaling problem. We just calculated the curl for a single, invisibly small microscopic loop. How do we scale this up to find the circulation of a massive loop of wire?

Imagine covering a large, macroscopic wire loop with a mesh of millions of microscopic squares (like a fishing net). If you calculate the tiny counter-clockwise circulation around every single square in the net, an amazing cancellation occurs.

Where two microscopic squares touch, their boundary lines are traversed in opposite directions. The circulation along all internal boundaries cancels out to zero! The only path that does *not* cancel out is the outer rim of the fishing net, because there are no adjacent squares there.

**The Theorem Statement:** The open surface integral of the curl of a vector field over any surface  $S$  is identically equal to the closed line integral (macroscopic circulation) of the field along the contour path  $L$  bounding that surface.

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_L \mathbf{A} \cdot d\mathbf{l} \quad (48)$$

**Why it works for exams:** If an exam asks you to calculate the difficult line integral of a complex vector field around a triangular or circular path, you can use Stokes' Theorem to convert it into a much simpler surface integral of the curl.

## 6 Phase 0 Summary and Exam Checklist

Congratulations! By reading through and understanding these derivations line-by-line, you have built the impregnable mathematical foundation required to dominate the rest of Electromagnetic Field Theory.

Before moving on to Phase 1 (Core Electrostatics), ensure you can confidently check off every item on this list:

- ☐ I can write out the differential length ( $d\mathbf{l}$ ) and differential volume ( $dv$ ) formulas for Cartesian, Cylindrical, and Spherical coordinate systems from memory.
- ☐ I understand the physical, real-world meaning of the Gradient (steepest slope), Divergence (sources/sinks), and Curl (vorticity/rotation).
- ☐ I can fully reproduce the step-by-step Taylor series derivation for Divergence in Cylindrical Coordinates. I understand why the area of the front face is larger than the back face ( $r d\phi$  vs  $(r + dr)d\phi$ ).
- ☐ I can fully reproduce the step-by-step 4-path derivation for the x-component of the Curl in Cartesian Coordinates. I understand why the baseline field terms cancel out and only the spatial derivative terms remain.
- ☐ I can write out the mathematical equations for the Divergence Theorem and Stokes' Theorem, and I can explain to a peer how "microscopic cancellation" leads to macroscopic boundary integrals.

Once this checklist is complete, you are fully prepared to tackle the physics of stationary charges in **Phase 1: Core Electrostatics**.